Quasars polarization in the context of the Habitable Worlds Observatory era.

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Introduction

Quasars are extremely luminous **active galactic nuclei (AGN)**, powered by accretion onto a **supermassive black hole (SMBH)** at the center of a massive galaxy. Their central engine is surrounded by emitting, absorbing, and scattering regions, producing radiation across the **electromagnetic spectrum** (Fig. 1).

■ **Why study quasars?** A unique laboratory to study **accretion physics**, SMBH growth, and the connection between AGN activity and **galaxy evolution**.

■ **How can we probe their structure?** **Spectroscopy** probes the physical properties of emitting regions, while **polarimetry** adds a geometric dimension. Using the **Stokes formalism**, we characterize polarization through:
→ **Polarization degree** (P): how strongly polarized is the light.
→ **Position angle** (θ): orientation of the polarization vector on the sky.

Together, these observables reveal structures that may remain hidden in total-intensity spectra. **Spectropolarimetry therefore probes quasar structure through both intensity and polarization.**

► **Our work**
Current instrumentation limits **high-resolution spectropolarimetry**. The proposed **POLLUX spectropolarimeter** on the **Habitable World Observatory (HWO)**, proposed by an European consortium [3], will open new opportunities to probe quasar structure and physics. Our goal is to explore the capabilities of HWO/POLLUX and the **instrumental requirements** needed to address these science cases. Two examples highlight the potential of POLLUX:

① **A binary SMBH in Markarian 231.** Testing whether spectropolarimetric signatures can constrain the **presence of a SMBH binary**.

② **Extreme Balmer line profiles.** Using polarization signatures to probe the **geometry on the internal emission regions** in AGNs with exceptionally broad and asymmetric Balmer lines.

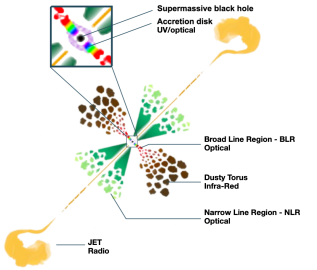


Figure 1: Schematic structure of an AGN [1]. The different components dominate at different wavelengths, providing complementary views of the same system. Adapted from [1].

① A binary SMBH in Markarian 231

Mrk 231 is a **quasar** presenting **distinctive UV-optical spectropolarimetric signatures**. Around ∼2500 Å, three independent observables show distinct and striking characteristics (Fig. 2):

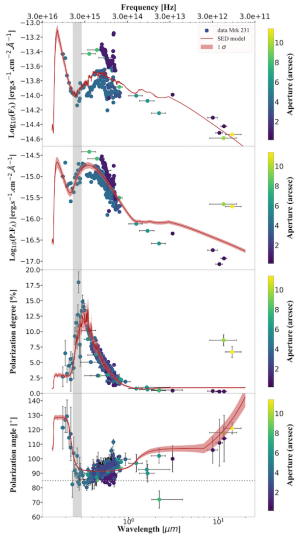


Figure 2: Results of the binary SMBH model fitting with Mrk 231 spectropolarimetric data [2].

- A sharp **drop in UV flux**
- A strongly **wavelength-dependent polarization degree**
- A smooth **rotation of the polarization position angle**

⇒ Could these signatures be explained by a **binary SMBH system**?

✔ We model the SED with four components: **two SMBHs, the host galaxy and the dusty torus**, and reconstruct their combined polarization signatures (Fig. 2).
A simple **binary SMBH model** simultaneously reproduces the independent measurements:

→ Total flux
→ Polarized flux
→ Polarization degree
→ Polarization position angle

➥ The observed polarization signatures, combined with radiative-transfer modeling, provide constraints on the **geometry of the binary system** (Fig. 3).

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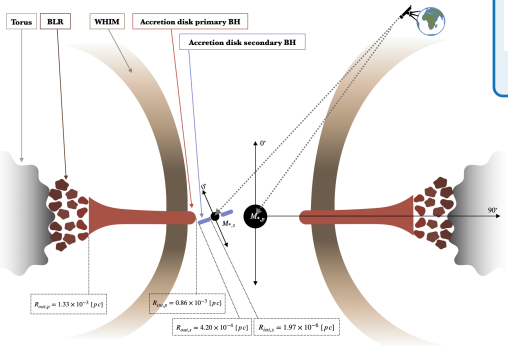


Figure 3: Schema of the structure of Mrk 231 based on the binary SMBH model, no to scale. [2]

② Extreme Balmer emission line profiles in AGNs

Some AGNs present **exceptionally broad and asymmetric Balmer line profiles**, reaching a width of several thousand km s^{-1} with corresponding unique polarimetric signatures (Fig. 4). Their unusual profiles and **complex spectropolarimetric signatures** provide a probe of the **inner geometry** of these sources.

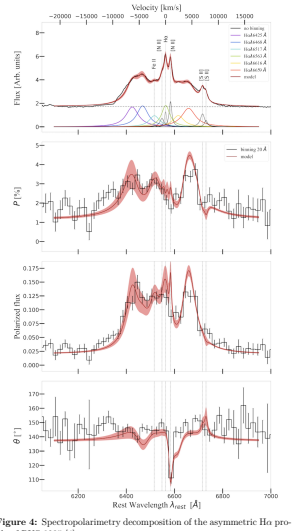


Figure 4: Spectropolarimetry decomposition of the asymmetric Hα profile of PKS 0235 [4].

- **Extremely broad** Balmer lines
- Strong **asymmetries and extended wings**
- Unusual **polarization degree variations** and **position-angle rotations**

⇒ What is the origin of these spectropolarimetric signatures?

✔ We decompose each Balmer-line profile into **kinematically distinct narrow and broad components** and reconstruct their individual polarization signatures (Fig. 4).

➥ The different components exhibit **distinct polarization signatures**, indicating that they do not share a single scattering geometry.

⇒ The observed polarization signatures constrain the **geometry of the emitting and scattering regions**, revealing a **stratified structure consistent with a disk wind** and connecting the **equatorial and polar scattering regions** (Fig. 5).

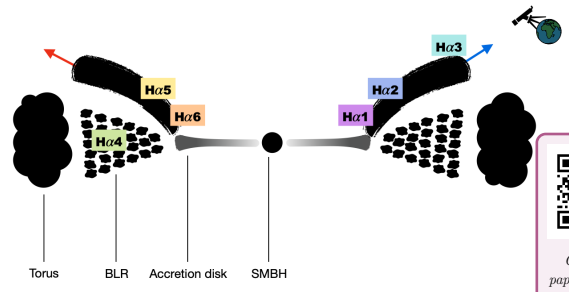


Figure 5: Schema of the structure of PKS 0235 based on the spectropolarimetric decomposition, no to scale.

Check the paper for more sources!

Conclusion

Spectropolarimetry provides a unique **geometric view of quasar structure**, revealing information that is difficult to access from total-intensity spectra alone.

Our science cases show how polarization can constrain the **geometry of SMBH binaries and the inner AGN regions**.

The high-resolution, broad-wavelength spectropolarimetry enabled by **HWO/POLLUX** could extend this approach to a wider range of quasar environments and physical scales.

References

[1] Marin, F. (2016), MNRAS, 460, 3679
[2] Biedermann, J., Marin, F., & Barnouin, T. (2025) A&A, 698, A107
[3] Neiner, C., et al. (2026) Astronomical Society of the Pacific Conference Series, 543, 267
[4] Biedermann, K., Marin, F., Hutsemékers, D., et al (2026), arXiv e-prints, arXiv:2609.29165.